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**INTERACTIVE 3D GRANITE CONFIGURATOR
WITH DESIGN QUALITY PREDICTION AND
QUOTATION MANAGEMENT SYSTEM FOR
SMALL TO MEDIUM ENTERPRISES**

**A Project Study
Presented to the Faculty of the
College of Computer Studies and Engineering**

In Partial Fulfillment of the Requirements
for the Degree of Bachelor of Science in Information Technology

by

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APPROVAL SHEET

The thesis, entitled: **INTERACTIVE 3D GRANITE CONFIGURATOR WITH DESIGN QUALITY PREDICTION AND QUOTATION MANAGEMENT SYSTEM FOR SIX SIGMAPHIL ENTERPRISES CORPORATION** prepared and submitted by **Christian Enayon, Dreizen Kyle Lopez, Keann Geoff Aguinaldo, and Prince Sazon**, in partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology** has been examined and recommended for acceptance and approval for Oral Presentation


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CERTIFICATE OF ORIGINALITY

We hereby declare that the thesis entitled, **INTERACTIVE 3D GRANITE CCONFIGURATOR WITH DESIGN QUALITY PREDICTION AND QUOTATION MANAGEMENT SYSTEM FOR SMALL TO MEDIUM ENTERPRISES** is our own original work carried out as Bachelor's student at Jose Rizal University except to the extent that assistance from others in the thesis' design and conception or in style, presentation and linguistic expression are duly acknowledged.

All sources used for the thesis have been fully and properly cited. It contains no material which to a substantial extent has been accepted for the award of any other degree at JRU or any other educational institution, except where due acknowledgement is made in the thesis.

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DATE

ABSTRACT

**Title : INTERACTIVE 3D GRANITE
CONFIGURATOR WITH DESIGN QUALITY PREDICTION AND QUOTATION
MANAGEMENT SYSTEM FOR SIX SIGMAPHIL ENTERPRISES
CORPORATION**

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The stone industry is filled with small and medium enterprises (SMEs), some businesses rely or still operate on manual processes, which can result in low scalability, inefficient quotation processing, and makes data-driven decision-making difficult. The researchers will develop a web-system for Six Sigmaphil Enterprises Corporation that will allow the customers to choose different project types and product designs to visualize the structures by

integrating pre-made 3D models from Meshy.AI. It also automates quotation prices depending on the user input. In addition, an admin dashboard module will be implemented to provide summary reports from system data such as popular product designs and popular project types. Furthermore, a traditional model will be implemented and evaluated for predicting the design quality scores of the stone designs.

Keywords: 3D granite configurator, quotation management system, design quality, predictive analytics, SMEs, data visualization, MeshyAI, R^2 , RMSE, MAE.

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Chapter 1 INTRODUCTION

This chapter presents the background, statement of the objectives, conceptual framework, significance, and scope and limitations of the study. The researchers will develop a web-based system for Six Sigmaphil Enterprises Corporation, a company working in the stone industry.

Background of the Study

The advancement of technology has significantly changed the way businesses operate, especially in terms of improving customer interaction, product visualizations, and decision making. The development of technology provides businesses to adopt modern technologies that can enhance efficiency and competitiveness to adapt in dynamic markets. Research has indicated that the implementation of such systems enables organizations to streamline operations and enhance the overall performance of businesses [1].

Despite these developments, many SMEs still face huge problems in utilizing information technology effectively. One of the problems these businesses face is the inadequate IT strategic planning, which can result in an imbalance between the objectives of the business and technological investments, leading to shortcomings and missed opportunities in growth [2].

Furthermore, SMEs usually experience limited resources, access in the market, and technological expertise.

Interactive visualization is a major technology that can make the customer experience better. Data visualization and interactive systems can greatly enhance users to comprehend and make better decisions by lessening cognitive load and enhancing the clarity of the product information [3]. Moreover, 3D visualization technologies enable the user to see a product more realistically, resulting in an improved customer engagement and purchasing decision.

Incorporation of data visualization is important in enhancing business operations. This allows businesses to turn small and large datasets into visuals like dashboards that are easier to understand. If it is implemented right, it can provide key insights into complex datasets in a way that can be practical and comprehensive [4].

The lack of a web system at Six Sigmaphil Enterprises Corporation can propose challenges in terms of customer consultation, visualizing products, and reports on trending products. Currently, the company uses platforms such as Viber, Facebook, and G-mail to communicate with customers; it can prevent the company from scaling to other potential customers as they have no software system to market their products. Furthermore [5], customers cannot visualize

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granite products in just images effectively which can lead to uncertainty in their purchasing.

Besides, the lack of structured analytics and data visualization can limit the capability of Six Sigmaphil to interpret trending products and make informed decisions. Dashboards and other visualization tools have been demonstrated to enhance decision-making by putting complex information in a form that can be easily understood and thus, businesses can determine trends and patterns better [6].

To address these challenges, there is a need to develop a web system that integrates pre-made, 3D granite models from Meshy that customers can configure, admin dashboard module for summary reports on trending products and designs, and a quotation management system to automate price estimates and sending quotations to the clients.

Statement of the Objectives

General Objective

The general objective of the study is to develop a 3D model configurator with design quality prediction and quotation management system for Six Sigmaphil Enterprises Corporation.

Specific Objectives

The present study will seek answers to the following objectives:

1. To design and develop a 3D granite configurator that enables users to adjust the size per sqm of the 3D model, and choose different granite designs, colors, cabinet finishes, and project types using 3D models generated from Meshy AI.
2. To develop a quotation management system that will automatically generate price estimates and quotations based on installation, labor, delivery costs, and customer inputs such as project type, granite design, and size per sqm.
3. To create, evaluate, and implement a predictive model using the dataset from COLOURlovers to predict the design quality score of the granite design chosen by the user based on design principles and project type suitability, and recommend compatible designs from the 16 available granite colors provided by Six Sigmaphil.
4. To implement an admin dashboard module that provides summary reports based on recorded quotation requests from users such as:
 - A. Popular product designs
 - B. Popular project types
5. To evaluate the proposed system using ISO 25010:2023 in terms of:
 - A. Functional suitability
 - B. Performance efficiency
 - C. Interaction capability

D. Security

Conceptual Framework

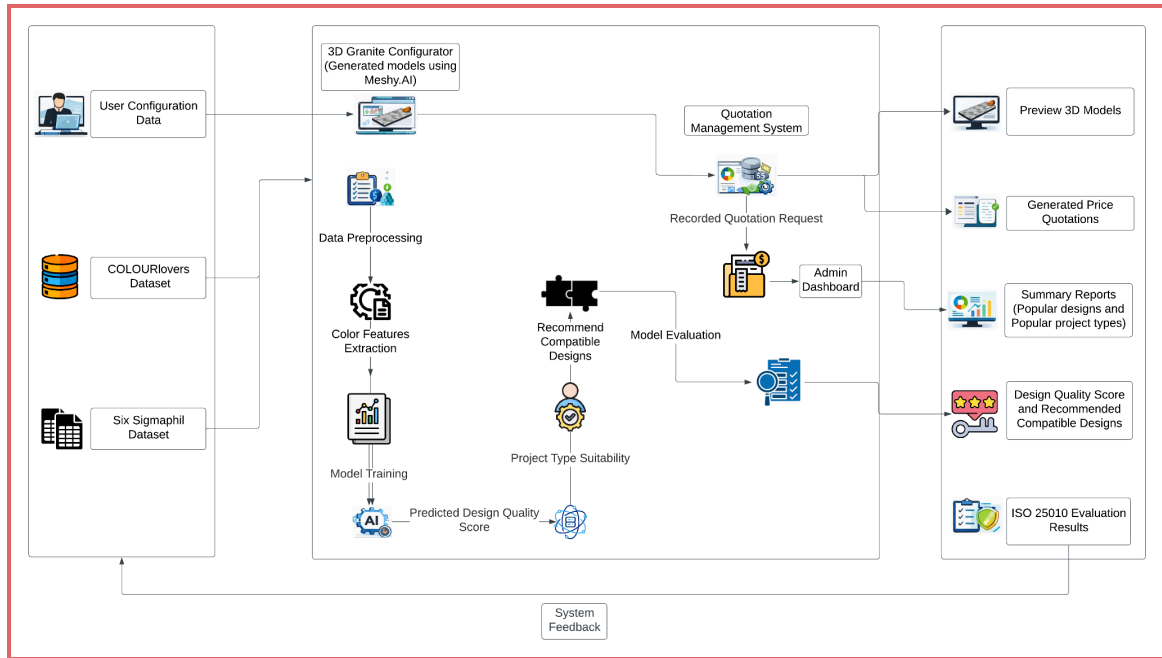


Figure 1.1 Research Paradigm

The conceptual framework of the study is based on the Input-Process-Output (IPO) Model which illustrates how the system transforms user configuration data, the COLOURlovers color palette dataset, and Six Sigmaphil's 16 granite designs into working outputs through four main processes, the 3D Granite Configurator using Meshy AI-generated models, a design quality prediction model, a Quotation Management System, and an Admin Dashboard providing summary reports such as popular product designs and project types, and with a system feedback loop for ISO 25010:2023 evaluation.

INPUT

User Configuration Data

These are the inputs that will be gathered during interaction with the system and include granite design, granite colour, cabinet finish, project type (e.g. kitchen countertop, bathroom countertop, wall cladding and flooring), 3D model adjustable size (length x width – measured in square meters). These inputs will be the foundation of the real-time 3D model preview, pricing estimate, and AI design quality score.

COLOURlovers Dataset

The data is publicly available, provided by O'Donovan, Agarwala and Hertzmann at SIGGRAPH 2011, University of Toronto [7]. It's 383,938 five-color palettes which are generated by the COLOURlovers online design community, each with its corresponding aesthetic quality rating assigned by people. This set of data will be used to train the predictive model to learn the complex color relationships and patterns that define the high or low rating that a set of colors will get when they come together visually in a well or poorly rated combination.

Six Sigmaphil Dataset

These are the 16 granite styles Six Sigmaphil offers (such as Calacatta Quartz, Blue Pearl, Black Marquina, Imperial Red, etc.) These will be the designs

that the trained model will evaluate and score on when a customer interacts with the 3D configurator.

PROCESS

3D Granite Configurator

Using the AI-based 3D model generator Meshy, the different types of structures like kitchen, bathroom, island countertops, wall and floor cladding will be created in a 3D form. These models will be displayed in the browser, and will be the main visual interface of the configurator, where the customer can instantly view various granites designs and cabinet finishes on their chosen structure.

Data Preprocessing

These raw COLOURlovers data will be loaded into a Python environment, and will be provided in a MATLAB .mat file format. The predictive model will evaluate the visual compatibility of two surfaces (granite design and cabinet finish), so only the first two predominant colors of each five-color palette will be extracted, giving a total of 383,938 2-color pairs. The extracted RGB color values will then be transformed to CIELAB (Lab) color space, which is a perceptually uniform colour space purposely designed to be closer to how the human eye perceives the difference in colour. The people's ratings will be normalized to a 0-100% design quality score so it will be a more intuitive number for the end user as they will be scaled up to a 0-100% score to a previously 0-5 rating.

Color Features Extraction

For each 2 color pair, 14 independent numerical features will be calculated from the values of their CIELAB colors. These will consist of the individual lightness (L), green-red axis (a), and blue-yellow axis (b) channels for both colors as well as derived comparative measures like chroma (color intensity), hue angle (color identity), and the mathematical differences between the two colors, Delta L (lightness contrast), Delta C (chroma contrast), Delta H (hue contrast), and Delta E (overall perceptual color distance). These numerical features will be the independent features that the model will be trained on.

Model Training

The preprocessed COLOURlovers data will be used for training a machine learning model. It will learn from the 383,938 pairs of colors and their quality scores, and find which color feature patterns of the CIELAB color space correlate to good and bad aesthetic quality.

Predicted Design Quality Score

One of the AI system's main functions will be to provide a design quality percentage for the granite and cabinet combination that the customer has chosen (e.g., 63%, 78%, 91%). The score will be calculated in real time, the colour characteristics (CIELAB) of the granite design and cabinet finish selected will be passed to the trained model to compute the score. The score will be computed

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on the aesthetic harmony of the two surfaces that is based on the patterns learned from the dataset.

Project Type Suitability

The evaluation of the AI model will be adjusted according to the project type the customer chooses. If cabinets are included in the structure, the granite colour will be compared to the colour of the selected cabinet finish, as in kitchen and bathroom countertops. For 3D structures without cabinets such as wall cladding and flooring, the model will evaluate and score the granite color against a white color, which simulates how the granite design would appear in a clean, modern space. This logic will help the AI score to be relevant to the particular application that the customer is designing for.

Recommend Compatible Designs

The machine learning model will score the remaining available granite design and cabinet finish against the customer's current selection, it will then sort them all by predicted design quality score, the highest scored granite design and cabinet finishes will be shown to the customer as recommended compatible designs. For structures with countertops, the granite recommendations will be scored against the selected cabinet, and cabinet recommendations will be scored against the selected granite. For the wall and floor structures, granite recommendations will show designs that best complement the customer's current granite design selection.

Model Evaluation

After training, the model will be tested with the traditional machine learning regression metrics such as R^2 (coefficient of determination that expresses how well the model explains variance in human ratings), RMSE (root mean squared error, shows the magnitude of prediction errors), and MAE (mean absolute error, shows the average prediction error). These metrics will help to ensure that the model has been trained with meaningful patterns from the data set before being deployed to real-time data.

Quotation Management System

Once the customer has configured their preferred design and structure this module will come in. The length and width of the granite design will be entered by the customer and the system will automatically calculate the estimated cost based on the granite design's price per square meter and the total area of the design will be calculated. The customer will then provide its contact information and the formal quotation will be added to the database for processing by the employees of Six Sigmaphil.

Recorded Quotation Request

Every request for a quotation will be logged and stored completely in the database. This will consist of the granite design, cabinet finish, project type, project dimensions, the calculated price estimate, and the customer's contact

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information. This data will be used for the Admin Dashboard's analytics and reports.

Admin Dashboard

This will be the back-end administration interface for use by authorized Six Sigmaphil staff only. It will offer them insights into all quotation transactions and graphs, allowing the team to understand the demand trends and make data-driven decisions and marketing.

OUTPUT

Preview 3D Models

The customer will be able to visualize the 3D model structure and their chosen granite design applied to their selected 3D model structure. The granite texture, finish on the cabinets and dimensions of the model will be updated immediately as the customer uses the configurator to see an accurate visual representation of the final product.

Generated Price Quotations

A formal quotation showing the material cost per sqm, total area, delivery cost, installation cost, discounts, and the total cost of the project based on the configuration inputs from the customer. The quotation will be automatically produced by the system once approved by the admin and the final quotation will be sent to the customer via Email.

Summary Reports

These are the reports that will appear on the Six Sigmaphil admin dashboard. The staff will be able to see the most chosen granite designs and project structures, and help the business to identify trends in demand and adjust their product lineup accordingly.

Quality Score and Recommended Compatible Designs

The AI-generated output that is shown to the customer in addition to the 3D model. This will encompass the design quality score of the granite and cabinet combination they have selected, a quick explanation of how the AI came to this score, and a ranked list of the top recommended cabinet finishes and granite alternatives including compatibility scores.

ISO 25010 Evaluation Results

The result of the assessment of the overall system according to the software quality ISO 25010:2023 standard. The evaluation will include functional suitability, performance efficiency, interaction capability and security, using a structured questionnaire filled in by the end users of the system and IT expert respondents.

Significance of the Study

The present study may be beneficial to the following:

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Six Sigmaphil Enterprises Corporation. The company may improve its operations and decision-making by having a web-based system

SMEs. The study may benefit SMEs by providing them a web-based system that may improve operational efficiency, customer experience, and data-driven decision making.

Customers. The study may be beneficial to customers by providing them product visualization. It may reduce uncertainty and improve customer satisfaction.

The Researchers. The study may improve the researcher's skills regarding system development.

The Future Researchers. The study may benefit the future researchers in developing similar systems.

Scope and Limitations of the Study

The study focuses on the development of the Interactive 3D Granite Configurator with Design Quality Prediction and Quotation Management System specifically designed for Six Sigmaphil Enterprises Corporation for pilot testing. The study is conducted within the operations of the company, which serves as the setting of the study, the participants of the study include homeowners, IT experts, and other business-related SMEs. The study will be conducted within Metro Manila only as it is the Philippine's center of businesses.

Limitations

The limitations of the study include the 3D granite configurator are only pre-defined 3D models created from Meshy AI, and only 16 designs provided by Six Sigmaphil.

Definition of Terms

Functional Suitability. Capability of the system to provide functions required of intended users when it is used.

Performance Efficiency. Capability of the system to perform its functions and be efficient in the use of resources.

Interaction Capability. Capability of the system to be interacted with by users to exchange information between a user and a system via the user interface.

Small to Medium Enterprises (SMEs). It refers to businesses that are independent and have a small number of employees and financial resources.

3D Granite Configurator. AI generated 3D models that allow the customer to choose different project types and select different granite designs.

Admin Dashboards. Stores the recorded quotations and provides summary reports such as, popular product designs and popular project types.

Quotation Management System. It provides a price estimate based on the customer inputs such as selected project type, product design, and size per sqm.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter presents the related literature and researches that are relevant to the study. Research and studies included supports the conceptual framework, system design, and implementation of the present study.

Related Literature

Local Literature

3D Product Configurator Visualization

Eyedex [8] explored what a 3D product configurator is and why it is essential. It discusses platforms integrating multimodal AI to understand text and images to create high quality products. It highlights the benefits of using 3D configurators such as boosted conversion rates, reduced return rates, increased average order value (AOV), and enhanced customer engagement.

Virto Commerce [9] discussed an overview of product configurators in B2B e-commerce, highlighting their capacity to minimize order errors and improve buyer confidence. The article highlights that product configurators can be enhanced by implementing 3D visualizations, this will ensure that the final order

matches the customer's expectations, therefore, it can reduce costly revisions and returns.

In the construction industry [10], product configurators facilitate effective data handling, reduce delivery time and resources, and provide better management of materials for the business with potentially measurable contributions to waste reduction and circular economy.

Additionally, Imagine.io [11] developed a B2B 3D configurator for customizing different products such as furniture, kitchen, and bath, which also has features like automatically generated bills, pricing, and PDF quotes.

Recent literature [12] states that 3D presentations are significant in enhancing online shopping experience through providing a more realistic, interactive, and immersive visuals of displaying products to consumers. The researchers highlight that unlike the traditional 2D display of products, 3D presentations provide users with the capacity of better visualization of the appearance, structure and feature of the product thus increasing the customer interest and buying intentions. It also addresses the technical aspects of successful implementation such as the correct input and output devices, and the interaction methods which would allow the users to manipulate product views. The review has revealed that using a number of case examples representing the

different business sectors, 3D visual technologies can have a positive effect on consumer behavior and even promote higher sales performance. The literature thus proposes that 3D presentation can be used to overcome the constraints of online shopping, specifically the lack of physical inspection with a product by developing a more immersive online experience.

According to D. Soukupova et al [13], the power of imagery and 3D presentations influences the customer's mental imagery, attitude, and purchase intentions. An analysis of expert studies also found that visual presentations, especially ones that show products, have a huge positive effect on customer engagement and purchase decisions. It was also found that the impact of visual elements varies by product type, which indicates that some products need better visual presentations than other products and that good visuals and clear description of the product help customers visualize the design better and make faster decisions even without physically seeing the product. Additionally, the research highlights businesses with references and guidelines for utilizing visuals to strategize in reducing customers uncertainty and influence purchase intentions.

According to C. Sun et al [14], discussed that online interactive visualization technologies considerably decrease the degree of product uncertainty among consumers. They highlighted that AR facilitates perceived

informativeness, presence, and visualization that subsequently decreases the

uncertainty on product quality and fit. This uncertainty minimization results in a higher number of positive product attitudes and better decision-making, which underscores the significance of interactive visualization in helping people to buy products online.

Foreign Literature

Mixed Reality Product Configurator

Lingyao Jin [15] proposed a Mixed Reality Product Configurator (MRPC) that allows users to customize their products in a mixed reality environment in an interactive manner. The system has features such as product modification, assembly, and real-time visualisation, which improves the levels of user involvement and satisfaction. It showed that configurators using MR prove to be better at customer experience by facilitating creative interaction, enhancing engagement, and helps in making more informed decisions. However, issues and challenges in adoption such as complexity in developing a system, user-acceptability, and cost which are essential variables to commercialisation of immersive technologies.

Random Forest Regression

Swetha et al [16] collected data on water quality parameters from the development of aquaculture (ADAK) to train and test the Random Forest

Regression model. Results show that the proposed model is competent in predicting the quality of water parameters accurately.

Wang et al. [17] created a design-process window architecture that uses machine learning to forecast and evaluate product quality in industrial polymer injection molding. To overcome the weakness of the conventional 2-D process window, which is limited by the parameters and cannot predict the quality response, the researchers used a pipeline of a Random Forest (RF) Classifier and Regressor to classify product defects and predict the quality response. The authors developed very descriptive and time-based features which are based on classical thermodynamic heat conduction functions to improve the predictive power of the models. Experimental results showed that by including these physically justified thermodynamic variables, the Random Forest model attained the 100% predictive accuracy rate.

Zavareh et al [18] discussed that Random Forest regression is efficient for predicting water quality indicators, especially dissolved oxygen, with land cover being a significant predictor, using ten years of data across 11 watersheds.

Predictive Modeling

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According to G. Ocean et al [19], they reveal that SMEs implementing predictive analytics have shown improvements in inventory management, revenue generation, and overall operational efficiency. Furthermore, businesses that utilize prescriptive analytics benefit from optimized resource allocation, enhanced marketing strategies, and better risk management practices. These show the advantages of AI to overcome the challenges SMEs face such as budget constraints and data availability.

Traditional forecasting methods often fail to adapt to dynamic market conditions, resulting in inaccurate predictions and operational inefficiencies [20]. To address these limitations, the study proposes the integration of predictive analytics techniques, including Linear Regression, ARIMA, Facebook Prophet, and Long Short-Term Memory (LSTM), to enhance forecasting accuracy by capturing both linear and non-linear patterns in sales data. The findings reveal that while Linear Regression provides a simple and interpretable baseline model, advanced techniques such as Prophet and LSTM demonstrate superior performance in handling seasonality and complex temporal dependencies, as evidenced by lower error metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Moreover, the study highlights the importance of deploying these models within an interactive system that allows real-time visualization, model comparison, and user-driven forecasting, thereby bridging the gap between technical modeling and practical business application.

According to Puppala [21], linear regression models establish a mathematical relationship between dependent and independent variables, enabling accurate forecasting and data-driven decision-making across various domains such as economics, healthcare, and business analytics. The study emphasizes that multiple linear regression is particularly suitable for real-world applications where multiple factors influence an outcome, making it highly relevant for systems that analyze sales performance and cost estimation. Furthermore, the effectiveness of regression models is evaluated using performance metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 , which measure the accuracy of predictions and the model's ability to explain variability in the data. The study also highlights limitations such as multicollinearity, sensitivity to outliers, and assumptions of linearity, which must be addressed through proper data preprocessing and model validation techniques.

Regression modeling enables the identification of complex and hidden relationships among quantitative variables, allowing organizations to transform raw data, such as sales transactions, system logs, and operational records, into useful and actionable information [22]. This transformation is particularly significant in the context of Big Data, where the volume, velocity, and variety of

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information requires robust analytical methods to uncover trends and patterns that are not immediately visible. Furthermore, regression analysis supports predictive capabilities by utilizing historical and current data to forecast future

organizational outcomes, including productivity, profitability, and overall performance. The study also emphasizes that both simple and multiple linear regression models are practical tools for analyzing real-world business scenarios, especially when multiple variables influence outcomes, making them highly applicable in performance evaluation and strategic planning.

Related Studies

Local Studies

3D Modeling

A study by S. Trivedi et al [23], they examined the issue of limited product visualization, the study showed that utilizing 3D models empowers customers to make well-informed decisions before making a purchase.

J. B. Davies et al [24] stated that implementation of AI tools that can generate 2D images to a 3D interactive model reduces the need for extensive technical expertise and significant time investment in creating visual content for product design.

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A study by J. M. G. Sanchez et al [25] shows the importance of a car configurator data systems in acquiring preferences of customers and forecasting future demand. The research analyzes the relationship between sales data and

user activity in car configurator webpages, results show that users visit car configurator webpages 1-6 months prior the purchase date. Additionally, the correlation between the car configurator and actual sales will allow car manufacturers to identify which specific color mix or car models customers are interested in prior 1 to 6 months purchasing.

According to M. Yi et al [26], results show the significance of user interface design and product customization. Furthermore, product pricing and navigation design of the product configurator was analyzed to determine the most suitable interaction design principles to enhance the user experience of product customization to provide an idea to the customers.

A comparative study by H. J. Kim et al [27] has been conducted between conventional e-commerce and 3D VR shopping, results show that virtual reality is effective, delivered deep immersion, and improved product information recognition. VR technology enabled users to see products in a 360° angle and view the products in detail leading to acquisition of a better understanding of the product than the traditional 2D images. In the online shopping context, the results

suggest that 3D visualization technologies are effective in enhancing the decision-making process of the customers and in lessening ambiguities related to online purchases.

Foreign Studies

Predictive Analytics

Based on a study by F. Ugbebor et al [28], the results demonstrate that predictive analytics models can improve decision-making and business performance. Forecasting models, specifically to market trends, can help SMEs plan in advance on possible changes in market trends and customer behavior. The study also highlights the potential benefits of predictive analytics for SMEs, such as improved operational efficiency, customer satisfaction, and competition. Additionally, the study indicates that there are some challenges with the use of business intelligence like data quality problems, lack of monetary capital, and skills.

Findings by E. Opoku et al [29], shows that despite the advancement of predictive analytics, there is still a gap in understanding its influence on SMEs' operational efficiency and revenue growth in the region Sub-Saharan Africa. The study investigates and sends 150 copies of questionnaires to SMEs in the cities Accra, Ghana, Lagos, and Nigeria using purposive sampling techniques, results

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showed that predictive analytics has a beneficial and huge impact on both operational efficiency and revenues growth of SMEs. Furthermore, the study recommends SMEs should give additional training to their staff to build the needed fundamentals to make the best of using predictive analytics.

W. K. Aguilar et al [30] indicated the effectiveness of utilizing predictive analytics in business processes within small enterprises, results show the benefits of a web-based data-driven analytics system using linear regression that is used to manage income and operations. The system automated sales tracking, inventory tracking, management of tasks and forecasting income based on historical data. The system was evaluated using ISO 25010 standards and showed that the system was highly usable, reliable, and performed, with the ability of predictive analytics established to enhance the decision-making process, operational efficiency, and financial planning.

According to J. Li [31], the study addresses the core factors of small-businesses weaknesses, examines the capacities and capabilities and constraints of predictive models, and evaluates how machine-learning methods can support decision-making without implying theoretical assumptions. Findings show that AI-driven forecasting significantly improves early-warning accuracy, reduces information asymmetry, and improves managerial responsiveness.

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M. Gafarov [32] stated that by using historical data and machine learning algorithms, it can forecast demand by using predictive analytics. The study highlights challenges such as data quality, integration technicalities, and real-time analysis on dynamic market conditions. Furthermore, the study delves into the best practices for applying predictive analytics to obtain data-driven

decision-making. It can provide a guide for businesses aligning predictive capacities with strategic goals in developing a more agile and competitive operational framework.

The study on [33] the effective implementation of predictive analytics, results show the effectiveness of churn prediction significantly depends on customer attributes and salesperson attributes. It indicates that predictive results have to be contextualized and aligned with user capabilities to produce significant business impact. Additionally, lack of knowledge among users such as sales personnel can crucially limit the advantages of predictive analytics.

According to Mustapha O. and Sithole T. [34], their study used different machine learning methods such as Random Forest Gradient Boosting Regression (XGBoost), Linear Regression, and Autoregressive Integrated Moving Average (ARIMA) were analyzed and utilized to measure the effectiveness of sales forecasting. Furthermore, it was indicated that businesses leveraging these predictive models can make data-driven decisions to improve

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efficiency, reduce costs, and increase profitability. Moreover, the study highlights the importance of selecting the most suitable algorithm for a given dataset and problems and the required proper tuning of the model and validation to output reliable results. Error metrics such as RMSE and MAE are significantly important to understand and

interpret to effectively evaluate and compare the performance of models. Each model was tested using 80-20% train-test split. Results show that Linear Regression had an RMSE = 16221.0, MAE = 12433.0, and an $R^2 = 0.990716$. ARIMA had an RMSE = 14959.8, MAE = 111265.3, and $R^2 = 0.983564$. Lastly, XGBoost resulted in an RMSE = 13574.7, MAE = 11649.6, and $R^2 = 0.993498$.

A research study by Yadav S. S. [35] discusses that sales prediction is an important factor in business operations and enables SMEs to make informed decision-making, minimizing risks, and leveraging opportunities in a fast-moving environment. Findings show in comparing multiple machine learning algorithms for SMEs, random forest gives the best results with an $R^2 = 0.61\%$, Linear Regression $R^2 = 0.76\%$, and XGBoost resulted with a $R^2 = 0.98\%$.

In the present competitive market [36], sales forecasting is important for business success. The researchers used a method that combines Support Vector Machines (SVM) and Artificial Neural Networks (ANNs) to discover potential

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customers and meaningful patterns. By utilizing machine learning techniques, businesses can perform data-driven decision-making to polish marketing strategies, enhanced customer targeting, and more personalized communication. The results show high precision with low error metrics (MAE = 0.29, MSE = 0.13, RMSE = 0.36, $R^2 = 0.99$), this shows the model's capability to handle complex

sales data. The study indicates that it improved marketing efficiency and the capability to excel in a competitive environment.

Ismail M. et al [37] on the need of predictive analytics to gather insights on consumer behavior for strategic sales due to the competitive supermarket industry. The study used multiple algorithms such as Random Forest, XGBoost, and support. The research performed in-depth data preprocessing and feature engineering, with the use of a stacking ensemble model, which resulted in strong predictive performance. The ensemble model achieved an $R^2 = 0.9990$ and low MAE. The findings show that machine learning techniques have the potential to improve sales forecasting and provide supermarkets with informed decision-making, optimized inventories, and being more focused.

The accuracy of sales forecasting plays a key role in strategic decision-making and resource allocation [38]. The results show a shift to advanced methods, an increase of adoption in machine learning techniques. This

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indicates the advancing prominence and effectiveness of handling datasets like time series. Furthermore, implementing effective and contextual factors such as seasonality, promotional activities, and market trends, is important for enhancing accuracy of predictivity. Also, handling interpretability and clarity is crucial for trust in the predictive models and incorporating formal decision-making in the automotive industry.

Quotation Management Systems

A study by M. A. H. Azizul et al [39] about a centralized quotation management system for quantity surveyors based on Excel and VBA indicated that the process of manually performing quotations is usually fragmented and can be prone to errors, which further indicates that the systematic implementation of digital tools is necessary. Nevertheless, the suggested system was not very scalable and could not be integrated with sophisticated analytics and innovative web technologies.

According to Mohammed Sharif and Anil Vasoya [40], their research in automating cost estimation and quotation in quantity surveying, replacing manual and excel-based systems with an ASP.NET-based system shows the efficiency and accuracy of the system was improved. The system had a 99% accuracy and shortened the time spent on estimations by 114 days to 3 days only, which demonstrates the usefulness of web-based automation in construction cost management. Moreover, the paper highlighted that SMEs have difficulty in

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embracing advanced systems because of cost and infrastructure constraints, which supported the necessity of low-priced scalable digital solutions.

Based on the study by A. G. Argomede et al [41] on the introduction of a web-based system that optimized the quotation process showed that the operational efficiency and customer service had improved significantly. The

system decreased the mean time of quotation generation to 28 minutes as compared to 55.87 minutes and the rate of compliance of submitted quotations to an average of 89.8. The results validate that web-based systems in the form of automation of quotation operations improve the response time rates, accuracy, and business performance.

Attribute Analysis and Modeling of Color Harmony

Wang et al [42] undertook a detailed research into quantifying color harmony by analyzing physical color features, including lightness, saturation and hue contrast, from colourful visual scenes in the real world. Their study evaluated different predictive performance models and found out which machine learning algorithm decodes the subjective human aesthetic assessment the best. The study showed that the Random Forest (RF) algorithm with 80.2% accuracy rate had the best performance in color harmony classification and prediction. This confirms that Random Forest regression could be used for assessing aesthetic

properties, and that complex multi-variable color features could be taken efficiently and successfully into account.

Color Prediction and Recommendation

Color is fundamental in automated design recommendation systems, particularly in terms of extracting and clustering color features. Yuan [43] applied state-of-the-art clustering models to process image features and generated

harmonious color schemes. On analyzing the current state of the practice on employing machine learning techniques to aesthetic design, the study found that dominant color extraction algorithms such as K-Means were common, while advanced color prediction and recommendation algorithms such as the Random Forest were widely used and highly effective. The results encourage the approach of using extracted color profiles to predict color model matchings to create design pairs that are compatible and pleasing.

Design Principles

Weisz et al. [44] presented a framework that is rooted in human interaction and includes six basic design principles and 24 actionable strategies specifically for computing applications that present AI-altered output to humans. The researchers identified two major modalities of use with AI workspaces: optimization, in which the user tweaks an existing artifact to satisfy specific task requirements; and exploration, in which the user uses the system to uncover

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possibilities within the domain and to capture inspiration. Importantly, the designing for imperfection principle points to the importance of using domain-specific measures to assess outputs as this directs user attention and speeds up the selection process when assessing visual or analytical artifacts, when evaluated against objective, measurable measures. Further, the framework defines boundaries, rationales, and feature parameters in advance and requires

systems to communicate them to users to set clear expectations and help reduce over-reliance on the automated judgments.

Thomas et al. [45] explored the effect of spatial dimensions and physical structural forms on players' perception, cognitive load, and visual engagement by converting a traditional planar gameboard into a volumetric multi-layered 3D structure. The empirical results indicated that a 2D plane gives way to a 3D spatial layout. The empirical findings showed that the spatial layout from a 2D plane is a 3D spatial layout which requires more cognitive demand from the user, in terms of spatial reasoning and dynamic strategic adjustments. To facilitate this shift, without inducing cognitive fatigue, the researchers illustrated the relevance of visual design principles. In particular, the study revealed that the implementation of Framing creates the appropriate physical structure and spatial clarity, and Contrast (through the use of clearly different visual and tactile materials) greatly decreases spatial confusion. In addition, significant positive

correlation was found between user engagement and the balance and framing principles ($r \approx 0.67$), indicating that the use of structurally balanced design elements in interactive interfaces has a direct effect on increasing aesthetic appeal, usability and user satisfaction.

Generative AI on 3D Modeling

Gallega & Sumi [46] studied the techniques, possibilities and visual impact of using generative artificial intelligence to generate and map realistic material textures in 3D interior design scenes. The study showed that generative texturing workflows are highly effective in improving visual fidelity and consistency in a 3D viewport. Moreover, this approach helps to reduce the threshold for end-users to explore devices with customizable digital space, providing immediate high fidelity visual feedback that closely resembles the look of physical materials in different spatial arrangements.

Kumar et al [47] discussed that common modeling techniques often demand huge manual input in areas such as modeling and texturing. To address these challenges, taking advantage of generative AI technologies in creating 3D models to enhance efficiency and streamline processes. By using the power of

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generative AI, it reduces the manual effort needed in creating high-quality 3D models, therefore speeding up the overall development timeline.

Gebreab et al [48] emphasized that Generative AI accelerates digital twin development by automating 3D modeling from 2D images and integrating data sources using 3D generative models and Large Language Models. Highlighting the potential of GenAI to minimize manual effort and streamline the integration of digital twins components. In particular, the researchers showed how it can

accelerate the creation of 3D models for DTs from 2D images by using 2D-to-3D generative models.

Synthesis of the Related Literature and Studies

The reviewed literature and studies indicate the evolution of business processes from disjointed to intelligent, unified, and web-based digital ecosystems around the world. The synergy between these was especially significant for Small and Medium Enterprises (SMEs), which seek to optimize their operational efficiency and customer interactions by seamlessly integrating interactive design tools with data-driven workflows [19, 28]. These unique domains are integrated to offer a comprehensive pathway to developing an ecosystem that can overcome the three challenges of physical examination, subjective design selection, and slow/manual transactional pipelines [13, 39].

Interactive 3D Visualization

The 2D product display methods are not sufficient to support online commerce because they increase the uncertainty of consumers, order mistakes and post-order corrections [9, 12, 14]. These problems can be directly solved by interactive 3D presentation technologies that can engage a consumer's positive mental imagery, boost purchase intentions, and provide data in the form of a picture that can give the business a 1-to-6-month forecasting window into consumer material preferences [13, 25, 27].

Configurators are used successfully in construction and interior design for optimization of data handling and the control of material overhead [9]. This is because traditional 3D pipelines have long relied heavily on manual processes and technical skills, whereas pipelines powered by generative AI like Meshy AI workflows are far more efficient, reduce the engagement of content creators, and provide immediate, high fidelity visual feedback [24, 46, 48]. Although the spatial cognition problem of relocating visitors from a flat 2D plane to a 3D layout has been mitigated, the introduction of interaction design principles is still crucial to limit spatial confusion and to optimize the spatial exploration of the visitor in 3D [26, 44, 45].

Predictive Analytics and Random Forest Regression.

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In addition to spatial visualization, the use of predictive analytics in small businesses contributes to increased sales and improves business decisions. [19, 29, 30] However, when linear models are not able to represent the non-linear structure of the data, modern machine learning techniques are used to analyze the data without making a prior theoretical assumption [20, 31]. To this end, the Random Forest (RF) algorithm has been found to be very capable of tackling such multi-variable relationships and generating strong and accurate predictions in various empirical fields [16, 17, 18, 34, 35]. Importantly, this predictive ability also applies to subjective human aesthetic judgments, as quantified. Empirical experiments show that the Random Forest models are able to predict human aesthetic judgements based upon physical color attributes such as lightness, saturation, and contrast of hues with a high success rate of 80.2% correct classifications of color harmony [42]. Moreover, the advanced color extraction and clustering algorithms such as K-Means with Random Forest models are very successful in predicting the compatibility of the material matching and providing design suggestions in harmony [43].

Quotation Management and SME operations.

Manual administration is a significant constraint for operational efficiency in client-facing environments. There are documented aspects of manual quotation systems, which are based on Excel or VBA, that are inefficient, error-prone and cannot scale to support growing customer volumes, which

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significantly slow the client response rate [39]. Web-based quotation automation is an efficient and cost-effective solution that meets the requirements of the infrastructure of SMEs [40]. By adopting an automated, centralized web architecture, organizations can attain up to 99% accuracy in cost estimates, and significantly shorten the estimation process from more than 100 days to just three days [40]. In conclusion, automation of quotation processes reduces average generation time by half and directly impacts on business performance and client response rate [41].

Relevance of Related Literature to the Current Study**Table 2.1** *Comparisons/Differences of Previous and Present Study***3D Modeling**

Title	Author	Date	Key Findings	Key Gaps	Present Study
3D Modeling: A New Tool for Improving the Online Shopping Experience	S. Trivedi et al	2024	Enhanced customization experience by providing AI-generated logos users can choose from and enhanced customer satisfaction	Not a small to medium enterprise level system.	The present study will include pre-made 3D AI models, quotation management system, and admin dashboards.
3D Model Generator: AI-Powered 2D-to-3D Reconstruction using Generative Models and Web-based Visualization	J. B. Davis	2025	Transforms 2D images to 3D models using Zero123 algorithm, generated outputs in .glb format for gaming, AR/VR, educational, and creative applications.	Focuses only on generating 3D models, no system for handling quotations, limited customization, and only transforms 2D images to 3D models.	The present study will utilize a granite configurator that is customizable (different colors and granite finishes), implements a system for handling transactions.
Influence of Car Configurator Webpage Data from Automotive Manufacturers on Car Sales by Means of Correlation and Forecasting	J. M. G. Sanchez et al	2022	Shows that positive Pearson Correlation Coefficients exist for all car models under analysis, users visit car configurator webpages 1 to 6 months before the purchase date.	Focuses mainly on data analysis and forecasting, did not find evidence of internet data in the form of visits to the automotive brand's CC webpages	The present study will integrate both 3D configurator and dashboards, focusing on the granite/construction industry.
Creating a Sustainable E-Commerce Environment: The Impact of	M. Yi et al	2022	Greatly contributes to both interface interaction design and customization, interaction design significantly impacts	Focuses mainly on UX design.	The present study will implement more than just design

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Product Configurator Interaction Design on Consumer Personalized Customization Experience			user satisfaction and purchase intention.		and product visualization.
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Predictive Analytics

PREDICTIVE ANALYTICS MODELS FOR SMES TO FORECAST MARKET TRENDS, CUSTOMER BEHAVIOR, AND POTENTIAL BUSINESS RISKS	F. Ugbebor et al	2024	Predictive analytics have the capability to notably enhance decision-making and performance of SMEs. Predictive models can effectively forecast market trends, customer behavior, and potential business risk.	Does not discuss on how predictive models can be implemented on web-systems	The present study will integrate design quality prediction and provide reports like popular product types and granite designs in a web-system
Predictive analytics, operational efficiency, and revenue growth in SMEs in Africa	E. Opoku	2024	predictive analytics has a beneficial and huge impact on both operational efficiency and revenues growth of SMEs	Does not apply predictive analytics on a software system	The present study will implement predictive analytics into a web system
AI-Enabled Predictive Analytics for U.S. Small-Business Resilience: A Policy-Neutral, Data-Driven Assessment	Jinyuan Lin	2026	The benefits of a web-based data-driven analytics system using linear regression that is used to manage income and operations. The system automated sales tracking, inventory tracking, management of tasks and forecasting income based on historical data.	Does not implement AI predictive capabilities on online platforms or systems	The present study will integrate predictive analytics along with a quotation management system

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Predictive Analytics for Sales Forecasting and Inventory Management	M. Gafarov	2024	The use of predictive analytics can accurately forecast sales and predict demand, improving business planning.	Emphasizes only on sales and inventory.	The present study will give reports on popular product types, granite designs, and automate quotations
Effective Implementation of Predictive Sales Analytics	J. Habel et al	2022	When predictive models are implemented properly, it can enhance decision-making and strategic targeting.	It focuses only on churn prediction.	The present study will implement a predictive model for design quality prediction

Quotation Management System

Centralized Quotation Management System for Quantity Surveyors	M. A. H. Azizul et al	2024	The centralized quotation management system utilizing Excel with VBA showed improvements in data management, traceability, and reduced duplication records, enhancing operation efficiency.	The system is limited to only Excel-VBA.	The present study will integrate a quotation management in a web system
Web-Based Automation for Accurate Cost Estimation and Quotation	B. S. M. Sharif and A. Vasoya	2025	The system had a 99% accuracy and shortened the time spent on estimations by 114 days to 3 days	The system only focused on estimated cost for construction.	The present study will expand beyond cost estimation, it will also integrate a frontend and backend system.
Implementation of a Web System to Optimize the Quotation Process in the Company KSF Representacion es EIRL, 2022	S. A. G. Argomedeo et al	2022	The system decreased the mean time of quotation generation to 28 minutes, compared to 55.87 minutes and the rate of compliance of submitted quotations to an average of 89.8.	The system does not integrate predictive modeling and is mainly a quotation process optimization.	The present study will integrate predictive modeling using a traditional model or technique regression algorithm.

Chapter 3

METHODOLOGY

Chapter 3 presents the research design, data sources, research locale, method or procedure, instrumentation and data collection, and data analysis tools used for the study.

Research Design

The study adopts qualitative with a developmental approach, it will be implemented by using the collection of user feedback to evaluate the system in terms of functional suitability, performance efficiency, and interaction capability. The developmental approach is to focus on the system's design, requirements, development, and evaluation. It will ensure that the system goes through stages such as requirements analysis, user design, iterative development, and deployment. The responses of the respondents will be collected using a structured questionnaire based on a Likert scale. The answers will be analyzed to understand the extent to which the system has fulfilled the needs and expectations of the users.

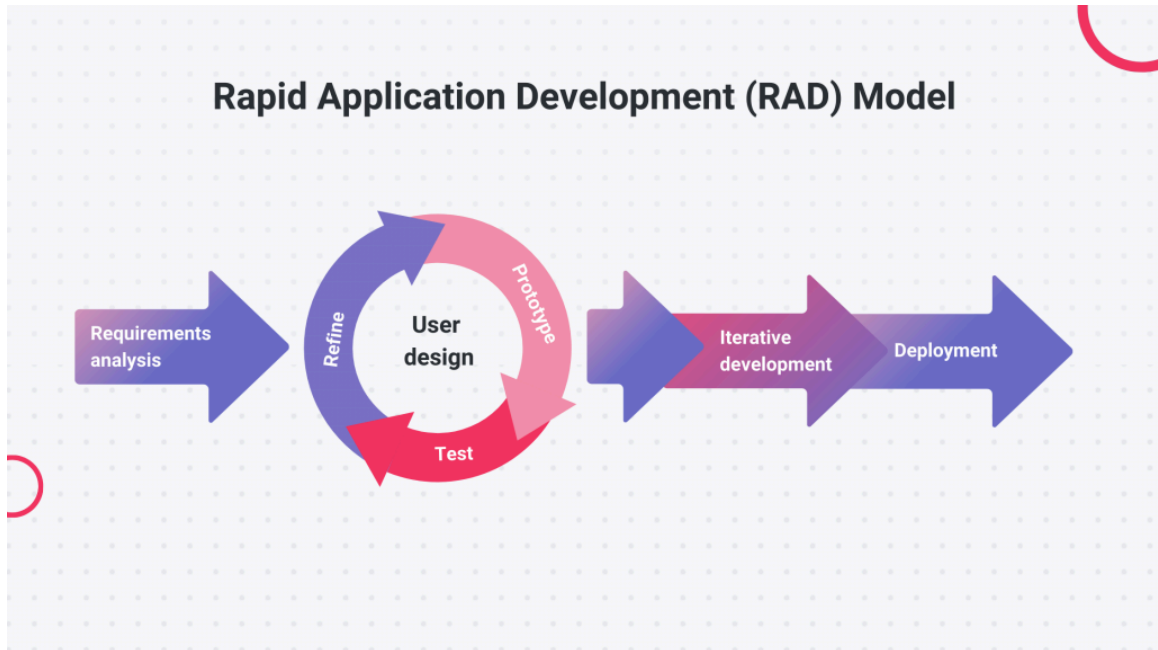


Figure 3.1 *Rapid Application Development [49]*

The researchers had chosen the Rapid Application Development (RAD) methodology as it is fast and flexible. In contrast to the more traditional methods like Waterfall, RAD allows the system to evolve through regular prototyping and receiving frequent feedback, therefore, the developers can refine the system functionality by revising the required features of the system to match the changing user requirements.

Requirements Analysis

This stage identifies the business requirements such as better product visualization by using 3D models, design quality score model, managing and sending quotations, dashboards for popular project types and designs.

The tech stack that the researchers will use for the development of the system are React and Tailwind for the frontend to create a responsive and user-friendly interface, with React Router to manage client-side navigation. Real time interactive 3D visualization was done using Three.js and React Three Fiber to display the AI pre-made 3D models made from Meshy AI. The system will implement Supabase as a Backend-as-a-Service (Baas), utilizing its managed PostgreSQL relational database for securing data persistence, Supabase Auth for identity management, and Supabase Storage for storing 3D .glb files and png/webp textures.

As opposed to the conventional methods which are very documentation intensive, RAD focuses on dynamic communication and flexibility which allows requirements to change as the project advances. Discussions are also held to make sure that all of the groups concerned have a common understanding of the purpose of the system.

User Design

During this stage, the developers will first create prototypes or mock-ups of the system. Such prototypes are not yet functional, but rather design and interactive depictions of the features of the system. This is aimed at providing the client to visualize the early versions of the system and give instant feedback. This

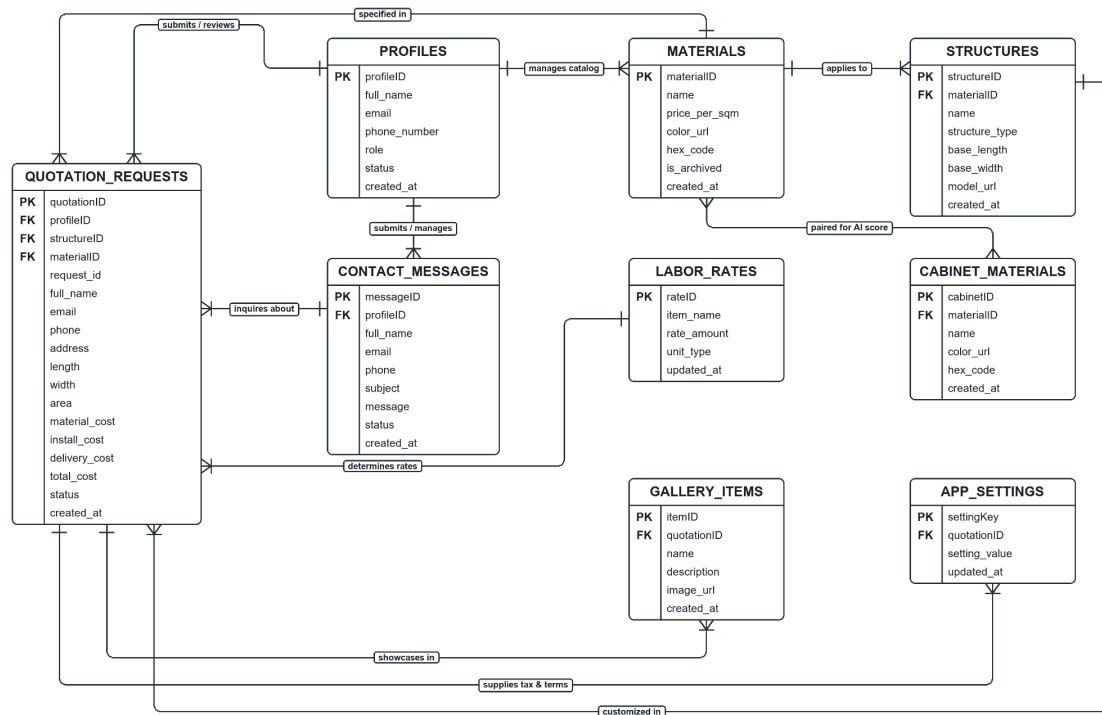
stage is iterative and thus the design is continuously developed as per the input of the users. This decreases the chances of significant revisions later in development.

Iterative Development

This stage entails the actual system development where prototypes will be converted to a working application. The developers are focused in coding, system integration, and implementing core features as they engage with the client for constant feedback. As opposed to the traditional models where testing is done at the end, RAD incorporates continuous testing and feedback loops thus enabling developers to make improvements without delaying the project.

Deployment

This is the last stage in which the system will be finally tested, quality assurance and deployment are performed. Before releasing the system to the client, the developers ensure that the application is evaluated in terms of functional suitability, performance efficiency, and interaction capability. Then, the system can only be deployed once the client approves. Additionally, RAD projects can still be improved upon even after deployment. Post-launch feedback is recommended to add future updates and changes to ensure that the system is still evolving post-release.



The Entity Relationship Diagram (ERD) manages the system where admins handle products, users submit inquiries, and quotations are generated based on granite selections.

PROFILES — It combines seamlessly with Supabase Authentication and brings together Administrators and Clients with RBAC. It saves the user credentials and user contact information (full_name, email, phone_number), account role (admin or user), and account status. Admins

can leverage this entity to view the pricing, catalog, and approve quotes, and clients use it to submit a request for quotation and monitor inquiries.

MATERIALS — The main stone product or catalog item (includes Granite, Engineered Quartz and Marble). It contains the name of the material, unit price per square meter (`price_per_sqm`), high resolution PBR texture file URL (`color_url`), dominant colour hex code (`hex_code`), and an archive status. It is the foundation for 3D visual representation, pricing algorithms and AI color compatibility assessments.

STRUCTURES — The 3D models in the configurator are stored here. Dimensional baselines (`base_length`, `base_width`) and structural categorization are stored in each record to enable users to upscale and view stone layouts in real-time 3D prior to requesting a quotation, without the need to download the 3D mesh model file (`model_url`) into their browser.

CABINET_MATERIALS — contains the available cabinets body and finishes (`name`, swatch image URL, and hex code). It is mainly designed to be used in conjunction with a stone material in the 3D configurator, sending two-color inputs to the model which outputs the design quality score.

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LABOR_RATES — is the pricing rates. It has standard commercial pricing for linear-meter stone cutting, exposed edge polishing, 45-degree mitering, site installation per square meter and delivery. It contributes by enabling company-wide pricing standardization, without having to hardcode in values for quotation calculation.

QUOTATION_REQUEST — is the main transactional object that integrates client quotations and official quotations into a single life cycle. It is created once the client finishes a 3D customization, and contains precise measurements (length, width, area), selected stone pattern, structure type and initial cost estimates, which are in a 'pending' state. Administrators then check the request and apply detailed fabrication charges (Cutting charge / Edge polishing charge / Miter charge), finalize any discounts and VAT, and send the final official PDF quotation to the inquiring party with the status marked as 'approved'.

CONTACT_MESSAGES — stores general customer enquiries and messages received via the public Contact page (as opposed to custom 3D quotes). It maintains all the sender's contact information, subject, message content and administrative status (unread, read, replied) and is an independent communication channel of customer service.

GALLERY_ITEMS — handles the firm's list of finished stone making and installing works. It captures project titles, descriptions, and media asset URLs, and helps inspire and gain customer trust by featuring verified Six Sigmaphil installations on the public gallery page.

APP_SETTINGS — is the configuration store for the entire system. It stores global operational settings, including the default Value-Added Tax (VAT) rate, quotation validity periods and payment terms, in structured key-value pairs, so that the financial logic is consistent throughout the platform.

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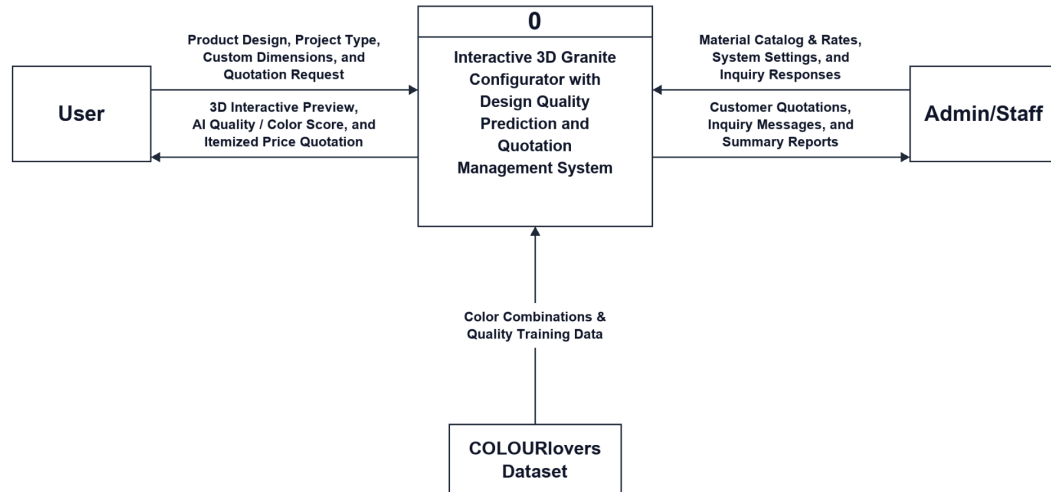


Figure 3.3 *Context Diagram of Web System*

Figure 3.1 illustrates the Context Diagram (DFD Level 0) of the Six Sigmaphil system, establishing the interactions between the central process and three external entities. Process 0 manages 3D structure configurations, AI color scoring, and quotation workflows. The User inputs custom dimensions, material preferences, and quotation requests, receiving interactive 3D visualizations, AI design quality ratings, and itemized pricing in return. Admin/Staff maintains material catalogs, labor rates, and system settings while responding to inquiries and reviewing incoming quote requests and summary analytics. Lastly, the COLOURlovers Dataset provides external color harmony data used to train the machine learning model for design quality evaluation.

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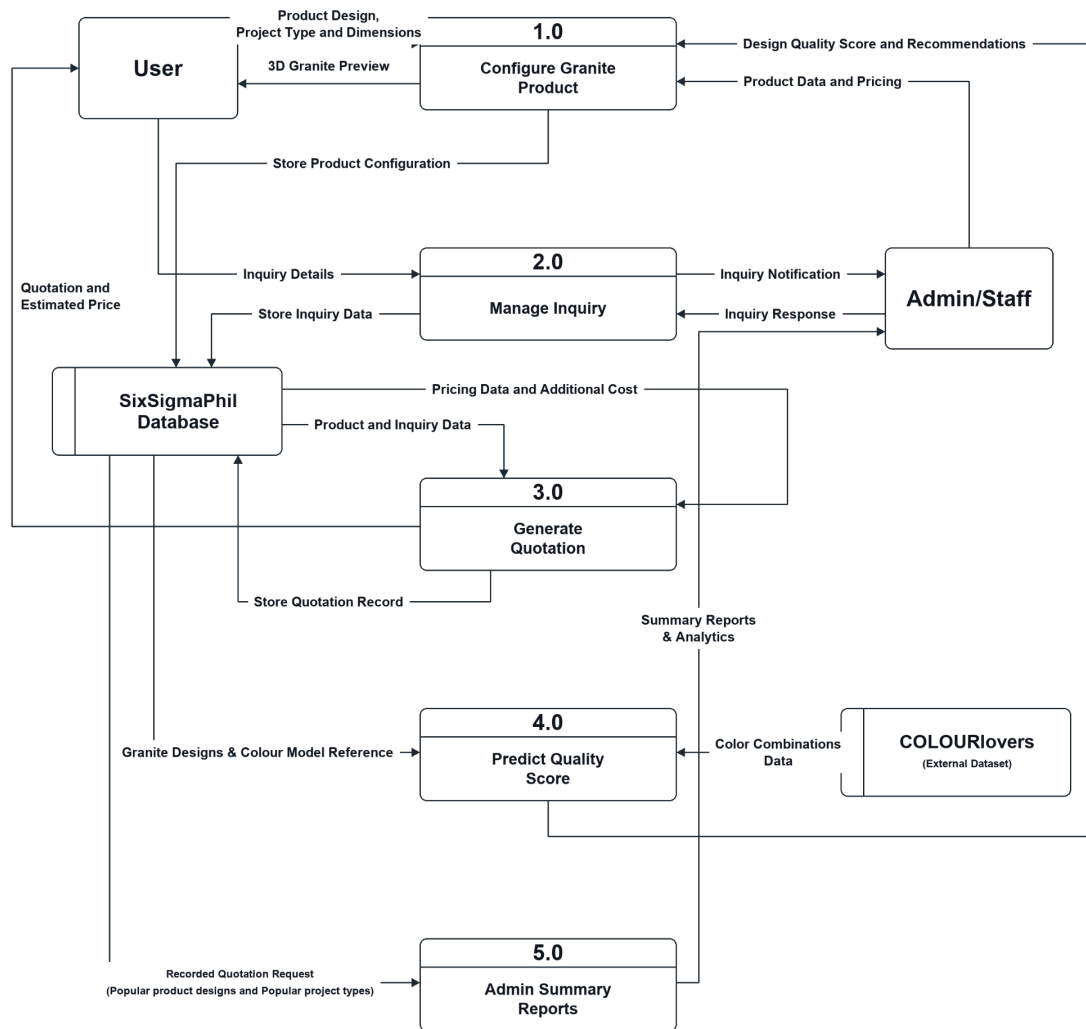


Figure 3.4 Level 1 Data Flow Diagram of Six Sigmaphil System

The level 1 Data Flow Diagram (DFD) gives more detailed information of the internal processes of the system and data flow between system components. It starts with:

Process 1.0: Configure Granite Product

Process 1.0 serves as the primary interactive module where clients visualize and customize their granite installations. Through this interface, users provide project specifications—such as project type, surface dimensions, and material choices—allowing the system to render an immediate 3D granite preview for real-time visual feedback. The process stores the finalized product configuration into the SixSigmaPhil Database for downstream processing. Additionally, it receives updated catalog specifications and material pricing directly from Admin/Staff, while seamlessly integrating AI-driven design quality scores and color harmony recommendations from Process 4.0 to help guide the client's design choices.

Process 2.0: Manage Inquiry

Process 2.0 facilitates structured pre-sales communication and customer service between prospective clients and company personnel. It captures user-submitted questions, custom requests, and project details, securely logging the inquiry records into the SixSigmaPhil Database for tracking. Upon receipt of a new inquiry, the process automatically dispatches an alert notification to Admin/Staff. Once administrative personnel formulate an inquiry response, the system relays the details back to the user, ensuring prompt consultation and client support.

Process 3.0: Generate Quotation

Process 3.0 (Generate Quotation) automates project cost estimation and formal quotation generation. The process retrieves the user's stored product configuration and inquiry data alongside active material unit rates, edge fabrication pricing, and additional service costs from the SixSigmaPhil Database. It computes the total estimated price, archives a timestamped quotation record within the database for auditing, and delivers an itemized quotation breakdown directly to the user.

Process 4.0: Predict Quality Score

Process 4.0 (Predict Quality Score) delivers automated aesthetic evaluation and color harmonization for configured products. It references internal granite catalog assets stored in the SixSigmaPhil Database and evaluates them against benchmark color palette relationships retrieved from the external COLOURlovers dataset. By analyzing color contrast, balance, and compatibility, the process generates a predictive design quality score and complementary recommendations, forwarding them to Process 1.0 to enhance the user's design experience.

Process 5.0: Admin Summary Reports

Process 5.0 (Admin Summary Reports) serves as the reporting and business intelligence module for company decision-makers. It extracts historical

quotation records, frequently requested project types, and top-performing granite product designs from the SixSigmaPhil Database. The process synthesizes this transactional data into comprehensive summary reports and analytics dashboards delivered directly to Admin/Staff, empowering management with actionable insights for inventory planning, design quality score prediction, and demand tracking.

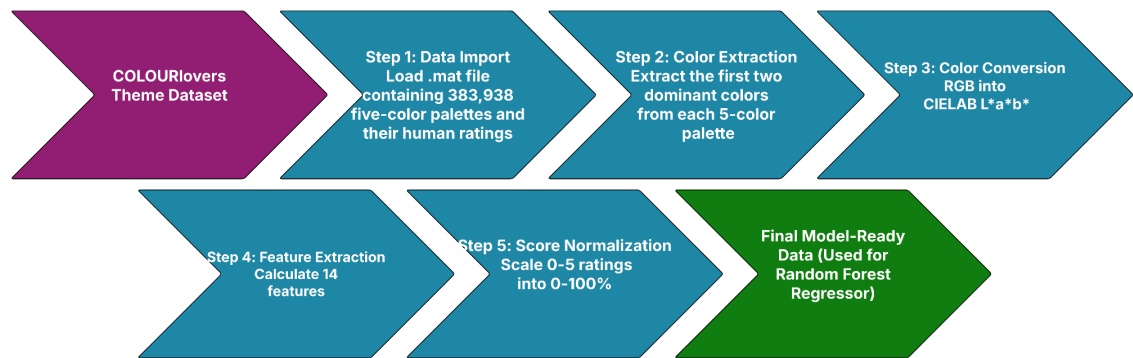


Figure 3.5 *Data Pre-Processing Pipeline*

COLOURlovers Theme Dataset

The data for this work comes from a public research data repository provided by O'Donovan, Agarwala and Hertzmann (ACM SIGGRAPH 2011, University of Toronto). It contains 383,938 five-color palettes by the COLOURlovers online design community, which have a corresponding aesthetic quality rating based on actual user engagement and preference data from COLOURlovers. This dataset was selected due to its extremely large size, with

almost 384,000 unique combinations of colors, reflecting a very broad and vivid spectrum of human color preferences.

Step 1: Data Import

The COLOURlovers dataset is loaded into a Python environment from a standard MATLAB .mat file format via the SciPy library. The two main arrays in this file are a data array of shape (383,938 x 5 x 3) with 383,938 user-made palettes of five colors each stored as RGB triplets, and a target array of shape (383,938 x 1) with the human-assigned popularity rating of each palette.

Step 2: Color Extraction

Because the predictive model has to compare the visual compatibility between two surfaces (granite countertop and cabinet color), the top two colors from each of five-color palettes are selected. This reduced the dataset from a 5 color structure, into 383, 938 distinct two-color pairs and each representing a realistic design pairing scenario.

Step 3: Color Conversion

The extracted RGB color values are converted back to CIELAB (Lab) color space. A conversion is necessary as CIELAB is a perceptually uniform colour space purposely designed to be closer to how the human eye perceives the

difference in colour, this makes it have a more accurate foundation for measuring aesthetic compatibility than just raw RGB values.

Step 4: Feature Extraction

In the CIELAB values, there are 14 independent variables calculated for each color pair. These include individual lightness (L), green-red axis (a), blue-yellow axis (b) channels of both the color 1 and 2, and derived metrics such as chroma, hue angle, and mathematical differences between the two colors, namely, Delta L (lightness contrast), Delta C (chroma contrast), Delta H (hue contrast), and Delta E (overall perceptual color distance).

Step 5: Score Normalization

The original COLOURlovers human rating (0-5 scale) are linearly normalized to a design quality score of 0-100% human rating. This change offers a more intuitive and user-friendly measure that can be easily understood by the customers in the UI interface. These 5 steps will result in a clean structured dataset of 383,938 rows, containing 14 feature columns and 1 target column which can be used to train the Random Forest Regressor model.

Population and Sample

The researchers will use a purposive sampling technique and used the Raosoft calculator to calculate the sample of this study, the respondents of this

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study will consist of 10 small to medium enterprises (SMEs) that are related to the businesses of Six Sigmaphil Enterprises Corporation. Moreover, the research will involve 3 IT experts that are in the industry for several years to review the technicalities of the system. Moreover, homeowners who are interested in products that Six Sigmaphil offers will participate as respondents to provide their insights.

Respondents	Total Number of Respondents	Target Respondents
SMEs	10	10
IT Experts	3	3
Homeowners	50	45
Total		58

Table 3.1 *Target Respondents*

Sources of Data

A. Dataset Description

The study is based on a large colour compatibility dataset, published by O'Donovan, Agarwala, and Hertzmann in their peer-reviewed paper "Color Compatibility From Large Datasets" (SIGGRAPH 2011, University of Toronto) and made public. There are 383,938 5-color palettes that were created by and rated by COLOURlovers online design community users. The large size of the data set was chosen to provide the machine learning model with a broad and

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wide variety of real-world human color preferences, with almost 384,000 unique human-rating combinations of colors. The samples are then preprocessed and used to select the top two most common colours per 5 colour theme, creating 383,938 training samples to represent a surface pairing (e.g. granite and cabinet). There are 14 mathematical color properties for each sample of two-color combination, plus normalized Design Quality Score.

Column Name	Description	Data Type	Example Value
L1	Lightness value of Color 1 in CIELAB color space (0 = black, 100 = white)	Numeric	72.4
a1	Green-Red axis value of Color 1 in CIELAB	Numeric	-5.2
b1	Blue-Yellow axis value of Color 1 in CIELAB	Numeric	18.6
L2	Lightness value of Color 2 in CIELAB color space	Numeric	34.1
a2	Green-Red axis value of Color 2 in CIELAB	Numeric	3.7
b2	Blue-Yellow axis value of Color 2 in CIELAB	Numeric	-2.1

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delta_L	Lightness contrast between the two colors	Numeric	38.3
chroma1	Vividness (saturation) level of Color 1	Numeric	19.3
Chroma2	Vividness (saturation) level of Color 2	Numeric	4.2
delta_C	Difference in vividness between the two colors	Numeric	15.1
hue_angle1	Position of Color 1 on the color wheel in degrees	Numeric	105.6
hue_angle2	Position of Color 2 on the color wheel in degrees	Numeric	330.2
delta_H	Angular difference between the two colors on the color wheel	Numeric	224.6
delta_E	Overall perceptual color distance calculated using the CIEDE2000 formula	Numeric	41.7
design_quality_score	Normalized average human aesthetic rating (0-100%), the dependent variable	Numeric	74.0

Table 3.2 *Data Characteristics and Description*

A. Dataset Characteristics

Origin and Collection Method

The data comes from COLOURlovers, a large social network where millions of designers, from professionals to amateurs, design, share and rate five-color palettes. Each palette is the result of the community's engagement, directly from the community's aesthetic judgement from real users. The predictive model learns from real people's ratings. This particular dataset, however, was carefully cleaned, curated, and validated with respect from the study by the researchers (O'Donovan, Agarwala, and Hertzmann) before it was published. The raw data was processed to reduce noise and provide statistically reliable results, then converted to a standard MATLAB format (.mat).

Color Representation

The RGB values contained in the dataset are numerical triplets on a scale of 0-1 representing the intensity of each of three colors: red, green and blue. These raw RGB values are subjected to a mathematical transformation to the CIELAB (Lab) colour space as part of the pre-processing. CIELAB is a perceptually uniform colour space purposely designed to be closer to how the human eye perceives the difference in colour. This transformation is a key component of the model's accuracy,

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making sure that the AI assesses color harmony and contrast based on the same perceptual standards as humans.

Feature Characteristics

Each record in the pre-processed dataset represents 2 color combination extracted from a 5 color palette theme. The 14 features computed reveal the underlying aspects of color relationships, lightness contrast (how light or dark each color is compared to the other, L and Delta L); chroma contrast (how saturated or not each color is, Delta C), and hue distance (how far apart each color is from the other on the color wheel, Delta H). It also includes Delta E, which calculates the overall perceptual color distance using the CIEDE2000 formula. Together, these features combined, gave the AI model the full mathematical description of each color combination, which it can then learn how to tell the difference between color combinations which are visually harmonious and those which are visually clashing.

Target Variable

The dependent variable in this study is computed from the original community ratings given in the dataset and was called the Design Quality Score. The raw ratings (0–5) are linearly normalized (0–100%). A higher rating, closer to 100%, equates to higher aesthetic appeal and visual

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harmony, while a score closer to 0% equates to a score that is seen as less aesthetically pleasing and harmonious.

Research Locale



Figure 3.6 *Six Sigmaphil Enterprises Corporation's Office*

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The study will be carried out in Six Sigmaphil Enterprises Corporation based in Mandaluyong City, Metro Manila, Philippines. Founded in 2008 by Roland Koh, their business is trusted and favored by many middle-class to high-end clients. The researchers chose the company because of their engagement in the supply, fabrication, installation, and expertise of high-quality granite, marble, limestone, sandstone, quartz, and slates.

The company was chosen as the study area because it is currently encountering the use of manual and messaging-based systems for handling customer inquiries, quotations, and product display by providing images or actual samples. The locale is important as it gives the study a real-world setting where the proposed system can be deployed, tested, and also evaluated. The requirements of the business are in line with the goals of creating a web-based system to reach out to more customers and scale their company.

Method or Procedure

The research process begins with the submission of the proposal to our technical adviser for review and approval. During this stage, the researchers will evaluate the feedback provided during the proposal consultation, refining and revising the manuscript accordingly to ensure that the study meets the expected academic standards before moving forward.

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Once the proposal is approved, the researchers will proceed to gather the business requirements, system needs, and constraints for the development of the system. This involves conducting interviews with Six Sigmaphil Enterprises Corporation to better understand the challenges, features, and the business requirements of the company. The insights gathered from these discussions will serve as the foundation for the subsequent system design and identification of key features that will need to be developed for the proposed system.

Instruments

Questionnaire: The study will use a structured questionnaire to assess the system using the ISO 25010:2023 standards, such as the functional suitability, performance efficiency, and interaction capability. The questionnaire will be handed to the selected respondents such as SMEs, IT experts, and homeowners to collect their insights.

Process Observation: The researchers observed the process in Six Sigmaphil Enterprises Corporation to gain insights about the different projects they work on, product designs, and how they communicate with their customers. The observations supported the researchers in determining the problems that exist to develop and design the proposed system.

Interview: The researchers interviewed the client to obtain information about their current processes, challenges, and system requirements. The interview involved the process in responding to customer inquiries, writing quotations, and processes.

Tools for Data Analysis

The researchers will apply frequency count, percentage, and weighted mean to discuss and interpret the data collected via the respondents. Performance metrics such as R^2 , RMSE, and MAE will also be used to evaluate the model.

For the specific objective 3, key metrics such as R^2 , RMSE, and MAE are used to evaluate the performance of the model to measure how good and accurate the model fits in the dataset. The researchers will use a traditional technique.

The model will be evaluated using:

R^2 - It will be used to indicate how well the model determines the variation in the target variable that is explained by the independent variables. It demonstrates the extent to which the independent variables such as price, area and project type contribute to the changes in revenue. It plays a role in defining the effectiveness of the model in capturing relationships in the data.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}$$

[50]

R ² Value	Interpretation
0.90 - 1.00	Very High Accuracy in design quality score prediction
0.70 - 0.89	High Accuracy in design quality score prediction
0.50 - 0.69	Moderate Accuracy in design quality score prediction
0.30 - 0.49	Low Accuracy in design quality score prediction
0.00 - 0.29	Very Low Accuracy in design quality score prediction

Table 3.3 *R² Interpretation*

Root Mean Squared Error (RMSE) - It will be used to measure the average size of prediction error, but it is focused more on larger errors. It plays a role in determining major variances between forecasted and actual revenues to ensure that the system reduces huge forecasting errors.

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

[51]

RMSE VALUE	INTERPRETATION
Close to 0	Very low prediction error in forecasting revenue
Small value	Low prediction error in forecasting revenue
Moderate value	Medium prediction error in forecasting revenue
High value	High prediction error in forecasting revenue

Table 3.4 *RMSE Interpretation*

Mean Absolute Error (MAE) – It will be used to measure the average difference between actual and predicted values. It is important in giving a comprehensible measure of prediction accuracy, enabling users to understand the amount of error to expect from the system's predictions.

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

[52]

MAE VALUE	INTERPRETATION
Close to 0	Very low average error in forecasting revenue
Small value	Low average prediction error in forecasting revenue
Moderate value	Medium average prediction error in forecasting revenue
Large value	Poor average prediction error in forecasting revenue

Table 3.5 *MAE Interpretation*

For the specific objective 5, frequency count, percentage, and weighted mean will be used in the analysis of the assessment of the developed system according to ISO 25010:2023 requirements such as functional suitability, performance efficiency, and interaction capability. Frequency will be used to count the quantity of responses, percentage will be used to demonstrate the percentage of the responses, and weighted mean will be used to find out the average rating of the system using the Likert scale.

- 1. Weighted Mean** - The weighted mean will be used to find out the average rating of the system using the Likert scale.
- 2. Frequency** - The frequency will be used to count the quantity of responses.

- 3. Percentage** - The percentage will be used to determine the percentage of the responses.

$$WM = \frac{\sum fx}{\sum f}$$

[53]

- WM = Weighted Mean
- f = frequency
- x = weight of each response

To interpret the weighted mean brought about by a 4-Point Likert scale, the following arbitrary scale will be employed:

Weight/Scale	Mean Range	Verbal Interpretation
4	3.26 – 4.00	Strongly Agree
3	2.51 – 3.25	Agree
2	1.76 – 2.50	Disagree
1	1.00 – 1.75	Strongly Disagree

Table 3.6 *Likert Scale Verbal Interpretation*

Chapter 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

Short introduction about chapter 4.

Chapter 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Short introduction about chapter 5

Summary

Have a summary aligned for each SOs.

Conclusions

Number of Conclusions equal the number of SOs.

Recommendations

Go back to your significance of the study and include those people in the recommendations.

State also to whom you are recommending the output of your study by citing it with the specific people mentioned above.

References

- [1] S. Nisrina, R. Masviansyah, A. Adiwana, I. Kusumasari, and R. Hidayat, "The Role of Information Systems in Improving the Efficiency of Business Decision Making," *J. Bus. Knowl. Dev.*, vol. 2, no. 2, Feb. 2025, [Online] Available:
<https://journal.pubmedia.id/index.php/jbkd/article/view/3470>
- [2] B. Molete, S. Mokhele, S. Ntombela, and B. Thango, "The Impact of IT Strategic Planning Process on SME Performance: A Systematic Review," *Smart Cities*, vol. 5, no. 1, Jan. 2025, [Online] Available:
<https://www.mdpi.com/2673-7116/5/1/2>
- [3] K. Eberhard, "The effects of visualization on judgment and decision-making: a systematic literature review," *Rev. Manag. Sci.*, vol. 73, pp. 167–214, Aug. 2021, [Online] Available:
<https://link.springer.com/article/10.1007/s11301-021-00235-8>
- [4] M. Islam and S. Jin, "An Overview of Data Visualization," Tashkent, Uzbekistan: IEEE, Feb. 2020, [Online] Available:
<https://ieeexplore.ieee.org/document/9012031>
- [5] TWIKIT, "What is the impact of poor product visualization on e-commerce profitability?" Twikit, 2025 [Online] Available:
<https://twikit.com/what-is-the-impact-of-poor-product-visualization-on-e-commerce-profitability/>
- [6] Scrubbed, "How Data Visualization Empowers Decision Makers," Scrubbed Blog, 2025 [Online] Available:
<https://scrubbed.net/blog/how-data-visualization-empowers-you-to-be-a-more-informed-decision-maker/>

- [7] <https://www.dgp.toronto.edu/~donovan/color/>
- [8] A. Furioso, "Best 3D Configurator Software 2025 Comparison," Eyedex Blog, Sep. 2025 [Online]
Available:
<https://eyedex.co/blog/best-3d-configurator-software-2025-comparison/>
- [9] Conquest, Marina "Product Configurator for B2C & B2B eCommerce: From Basics to 3D, Virto Commerce, 2025" [Online]
Available:
<https://virtocommerce.com/blog/ecommerce-product-configurator#tldr>
- [10] S. Shafiee, P. Piroozfar, L.-F. Forberg, H. N. Hansen, and E. Farr, "Enhancing efficiency and sustainability in construction: a product configurator for customizable off-site building solutions", University of Brighton, 2025 [Online]
Available:
<https://research.brighton.ac.uk/en/publications/enhancing-efficiency-and-sustainability-in-construction-a-product/>
- [11] A. A. Brooks "launches advanced 3D tool for furniture, building materials industries", *Furniture Today*, October 2025 [Online]
Available:
<https://www.furnituretoday.com/technology/imagine-io-launches-advanced-3d-tool-for-furniture-building-materials-industries/>
- [12] H. C. P. Vasquez "Importance of 3D presentation in the consumer's purchase intention", Vol. 2 (2022): DecisionTech Review [Online]
Available:
<https://dtr.pro-metrics.org/index.php/d/article/view/6>
- [13] D. Soukupova and P. Bachmann "The Power of Image: The Impact of Visuals on Online Consumer Decision Making" Publisher: Technical University of Liberec, 2025 [Online]
Available:
<https://dspace.tul.cz/server/api/core/bitstreams/4adaee97-227b-430b-ad10-1fe82fdf4625/content?authentication-token=eyJhbGciOiJIUzI1NiJ9.eyJ1aWQiOiJl5YjAwMzA5NC1kNzc1LTQwMWEtOGQ5YS05ZTdhN2QyNTEzZWYiLCJzZyI6W10sImF1dGhbnRpY2F0aW9uTWV0aG9kljoic2hpYmJvbG>

[V0aClsmV4cCl6MTc1OTc0MTI2MX0.XHK5wpVJu2tlkcEUKiF%5C_9aeuO2Oc%5C_HpNgZnLaGbOoSE](#)

- [14] Y. Fang, C. Sun, M. Kong, X. Chen, and Y. Liu, "Influence of augmented reality product display on consumers' product attitudes: A product uncertainty reduction perspective," J. Retail. Consum. Serv., vol. 64, Jan. 2022. [Online]
Available:
<https://www.sciencedirect.com/science/article/abs/pii/S0969698921003945>

- [15] L. Jin, "Adopting Mixed Reality Product Customisation in Brick-and-Mortar Retail: Stakeholder Insights for Commercialisation Challenge," pp. 1886-1899, Jul. 2025, [Online]
Available:
<https://dl.acm.org/doi/10.1145/3715336.3735750>

- [16] Random Forest Regression based Water Quality Prediction for Smart Aquaculture
Available:
<https://ieeexplore.ieee.org/document/10127488>

- [17] A Novel Sensing Feature Extraction Based on Mold Temperature and Melt Pressure for Plastic Injection Molding Quality Assessment
Available:
<https://ieeexplore.ieee.org/document/10057204>

- [18] Assessing the Efficiency of a Random Forest Regression Model for Estimating Water Quality Indicators
https://www.researchgate.net/publication/378273423_Assessing_the_Efficiency_of_a_Random_Forest_Regression_Model_for_Estimating_Water_Quality_Indicators

- [19] Ocran, G., Yusuf, S., O., Owusu, P., "AI-driven business analytics for SMES: Unlocking value through predictive and prescriptive analytic" International Journal of Science and Research Archive, 2024, 13(01), 3009–3022. October, 2024, [Online]
Available:
<https://ijsra.net/content/ai-driven-business-analytics-smes-unlocking-value-through-predictive-and-prescriptive>

- [20] Reddy, K., M., Ravikanth, K., Penjarla, N., K., "Sales Forecasting using Predictive Analytics: A Machine Learning and Time-Series Approach" Location: India, Publisher: International Journal of Research Publication and Reviews, Vol 6, Issue 8, pp 5476-5483 August, 2025, [Online]
Available:

<https://ijrpr.com/uploads/V6ISSUE8/IJRPR52240.pdf>

- [21] Puppala, A., "A COMPREHENSIVE REVIEW OF LINEAR REGRESSION MODELS: THEORY, APPLICATIONS, AND ADVANCED TECHNIQUES" Volume 3, Issue 1, January-June 2025, [Online]
Available:
https://doi.org/10.34218/IJMLC_03_01_001
- [22] Kalaian, S., Kasim, R., Kasim, N., "A Conceptual and Pragmatic Review of Regression Analysis for Predictive Analytics", 2017 [Online]
Available:
<https://www.igi-global.com/gateway/chapter/176831>
- [23] S. Tivedi, D. Kaushik, D. Saxena, and L. Somvanshi, "3D Modeling: A New Tool for Improving the Online Shopping Experience," India: IEEE, 2024. [Online]
Available:
<https://ieeexplore.ieee.org/document/10489098>
- [24] M, J. B. Davis E, A. K. Y, M. J. B, and M. P. Rampriya, "3D Model Generator: AI-Powered 2D-to-3D Reconstruction using Generative Models and Web-based Visualization," in *Proc. 4th Int. Conf. Applied Artificial Intelligence and Computing (ICAAIC)*, Salem, India, 2025, [Online]
Available:
<https://ieeexplore.ieee.org/document/11330616>
- [25] J. Sanchez, X. Cardona, and A. Martin, "Influence of Car Configurator Webpage Data from Automotive Manufacturers on Car Sales by Means of Correlation and Forecasting," *Forecasting*, vol. 4, no. 3, Jul. 2022. [Online]
Available:
<https://www.mdpi.com/2571-9394/4/3/34>
- [26] Z. Huang, M. Yi, and Y. Yu, "Creating a Sustainable E-Commerce Environment: The Impact of Product Configurator Interaction Design on Consumer Personalized Customization Experience," *Sustainability*, vol. 14, no. 23, Nov. 2022, [Online]
Available:
<https://www.mdpi.com/2071-1050/14/23/15903>
- [27] H. Kim, S. Jeong, and S. Kim, "Comparative Analysis of Product Information Provision Methods: Traditional E-Commerce vs. 3D VR Shopping," *Appl. Sci.*, vol. 15, no. 4, Feb. 2025, [Online]
Available:
<https://www.mdpi.com/2076-3417/15/4/2089>

- [28] S. Adeteye, F. Ugbebor, and J. Ugbebor, "PREDICTIVE ANALYTICS MODELS FOR SMES TO FORECAST MARKET TRENDS, CUSTOMER BEHAVIOR, AND POTENTIAL BUSINESS RISKS," *J. Knowl. Learn. Sci. Technol.*, vol. 3, no. 3, pp. 355-381, 2024, [Online]
Available:
<https://jklst.org/index.php/home/article/view/267>
- [29] E. Opoku, A. Kossidze, and O. Lawal, "Predictive analytics, operational efficiency, and revenue growth in SMEs in Africa," *World J. Adv. Res. Rev.*, Sep. 2024. [Online]
Available:
<https://wjarr.com/content/predictive-analytics-operational-efficiency-and-revenue-growth-smes-africa>
- [30] A. Vega, W. Aguilar, and R. Sidayon, "A Web-Based Data Driven Analytics System for Income and Operations Management Using Linear Regression for Modern Concept Prints," *Int. J. Res. Innov. Appl. Sci.*, vol. 10, no. 10, pp. 784-795, Nov. 2025, [Online]
Available:
[10.51584/IJRIAS.2025.1010000062](https://doi.org/10.51584/IJRIAS.2025.1010000062)
- [31] Li, J., "AI-Enabled Predictive Analytics for U.S. Small-Business Resilience: A Policy-Neutral, Data-Driven Assessment", *Econ. Bus. Rev.*, vol. 2, no. 1, pp. 18-24, 2026 [Online]
Available:
https://econpapers.repec.org/article/dbaejbema/v_3a2_3ay_3a2026_3ai_3a1_3ap_3a18-24.htm
- [32] Gafarov, M., "Predictive Analytics for Sales Forecasting and Inventory Management", *Global Res. Innov. Publ.*, 2025, [Online]
Available:
<https://globalresearchandinnovationpublications.com/NGJFTYR/article/view/83>
- [33] Habel, J., Alavi, S., Heinitz, N., "Effective Implementation of Predictive Sales Analytics", Volume 6, Issue 4, December 2022, [Online]
Available:
<https://journals.sagepub.com/doi/10.1177/00222437221151039>
- [34] Mustapha, O., O., Sithole, T., "Forecasting Retail Sales using Machine Learning Models", April 2025, [Online]
Available:
https://www.researchgate.net/publication/390949992_Forecasting_Retail_Sales_using_Machine_Learning_Models

- [35] Yadav, S., S., "Sales Forecasting for Small Medium Enterprises Using Machine Learning", Location: National College of Ireland, 2025 [Online]
Available:
<https://norma.ncirl.ie/7291/>

- [36] Negi, S., Tomar, V., Sharma, T., Kumar, A., "AI-Based Sales Forecasting Model for Digital Marketing", Volume/Issue: 11/2 | Page No: 1063-1070

2026, [Online]
Available:
<https://rsisinternational.org/journals/ijrias/view/ai-based-sales-forecasting-model-for-digital-marketing>

- [37] Ismail, M., Tukur, H., M., Friday, M., "Sales Prediction using Ensemble Machine Learning Model", Vol. 4 No. 3 (2025), [Online]
Available:
<https://www.ijsrmt.com/index.php/ijsrmt/article/view/350>

- [38] Ahaggach, M., Abrouk, L., Lebon, E., "Systematic Mapping Study of Sales Forecasting: Methods, Trends, and Future Directions", Volume 6 Issue 3, July 2024 [Online]
Available:
<https://www.mdpi.com/2571-9394/6/3/28>

- [39] Azizul M., H., A., Saini, M., N., Z., Nee, T., Y., "Centralized Quotation Management System for Quantity Surveyors", *Int. J. Res. Innov. Appl. Sci.*, Volume/Issue: 11/2, 2025. [Online]
Available:
<https://rsisinternational.org/journals/ijrias/view/centralized-quotation-management-system-for-quantity-surveyors>

- [40] B. Sharif and A. Vasoya, "Web-Based Automation for Accurate Cost Estimation and Quotation," *Int. J. Adv. Res. Comput. Commun. Eng.*, vol. 14, no. 4, Apr. 2025, [Online]
Available:
DOI: 10.17148/IJARCCE.2025.14423

- [41] B. De la Cruz, S. Argomedo, and L. Cabanillas, "Implementation of a Web System to Optimize the Quotation Process in the Company KSF Representaciones EIRL," *Int. J. Adv. Comput. Sci. Appl.*, vol. 16, no. 3, 2025. [Online]
Available:
<https://dx.doi.org/10.14569/IJACSA.2025.0160312>

JOSÉ RIZAL UNIVERSITY COLLEGE DIVISION

- [42] Attribute analysis and modeling of color harmony based on multi-color feature extraction in real-life scenes
<https://pubmed.ncbi.nlm.nih.gov/36186330/>
- [43] Generation of Graphic Design Color Schemes Based on CMYK Color Model and Corrosion Algorithms
<https://www.intmaritimeengineering.org/index.php/ijme/article/view/1397>
- [44] Toward General Design Principles for Generative AI Applications
<https://arxiv.org/abs/2301.05578>
- [45] Using design principles as a lens to interpret user needs: a case study using a new 3D tic-tac-toe game
<https://www.cambridge.org/core/journals/proceedings-of-the-design-society/article/using-design-principles-as-a-lens-to-interpret-user-needs-a-case-study-using-a-new-3d-tictactoe-game/729168008F58AEDB6384CF4895356266>
- [46] Exploring the use of generative AI for material texturing in 3D interior design spaces
<https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2024.1493937/full>
- [47] Implementing Generative AI into Realtime Visualization Workflow
<https://saemobilus.sae.org/papers/implementing-generative-ai-realtime-visualization-workflow-2025-28-0324>
- [48] Accelerating Digital Twin Development With Generative AI: A Framework for 3D Modeling and Data Integration
<https://ieeexplore.ieee.org/document/10786996>
- [49] Laptick, S., “The Power of Rapid Application Development: When Speed Meets Flexibility”, XBSoftware, November 2021 [Online]
Available:
<https://xbsoftware.com/blog/rapid-application-development-rad/>
- [50] Wikipedia “Coefficient of determination” [Online]
Available:
https://en.wikipedia.org/wiki/Coefficient_of_determination
- [51] Jacobi, O., “Root Mean Square Error (RMSE): The cornerstone for evaluating regression models”, August 2023 [Online]
Available:

JOSÉ RIZAL UNIVERSITY COLLEGE DIVISION

<https://coralogix.com/ai-blog/root-mean-square-error-rmse-the-cornerstone-for-evaluating-regression-models/>

- [52] datacamp “Mean Absolute Error Explained: Measuring Model Accuracy”
August 2025, [Online]
Available:
<https://www.datacamp.com/tutorial/mean-absolute-error>
- [53] Weighted Mean - Formula, CUEMATH, [Online]
<https://www.cuemath.com/weighted-mean-formula/>

Appendix A - Interview Transcript

1. What is the standard process when a customer reaches out or is interested in your products?

- We introduce the company and the product lines that we carry by giving them our Facebook profile. Then we ask the details of the project for their stone requirements by asking if they have an approved or specific color or type of stone and if they have drawings, plans and specifications. If there is none, we offer to conduct an actual measurement of the area to be covered with our stone products. We also offer to bring actual samples of the different kinds and colors of stone for the customers to choose from.

2. How do customers reach out?

- Viber, Facebook, referrals from existing regular customers, architects, engineers, and interior designers.

3. Who handles the initial response?

- The general manager or the sales assistant in the absence of the general manager.

4. How do you currently present or show product designs to the customers? Do you use pictures, or something else?

- We use pictures and videos of the different product colors and designs. We also present or send actual samples. Sometimes we invite the

customers over to our warehouse to see and inspect the actual slabs of stones.

5. How do you currently store records?

- Excel, Google Sheets, Viber, E-mail, and Hard copies.

6. What are the different product types you offer?

- Granite, marble, quartz, and sintered stones for kitchen countertops, bathroom countertops, island countertops, tabletops, floorings, wall claddings, mausoleums, columbariums, crypts, markers, and niches.

7. What are the charges that will be included in a quotation?

- Material cost, labor cost such as fabrication (cutting, polishing, moulding/edge profiling), installation cost, delivery cost, mobilization cost, and value added tax.

8. How do you currently create and send quotations to customers?

- Created in Excel or Google Sheets and sent through Viber or E-mail, sometimes we bring the hard copy directly to the customer.

9. How long does it usually take from a customer inquiry to a final quotation?

- If details are available such as plans, drawings and specifications (type, color and design of the stone) and actual measurements, around 2-3 days depending on the volume of the requirement or depending on the size of

the project. If some details are missing or not available, it could take a few days more to get the unavailable details such as actual measurements

10. How do you store past records?

- Excel, Google Sheets, Viber, and sometimes hard copies.

11. Are there any challenges you currently face in your operations?

- The lack of manpower when there are many projects at the same time.

12. Are customers sometimes uncertain in picking designs because of the lack of visualization?

- Yes, but most of the time, they can easily choose from the pictures, videos, or actual samples.

13. Would a software system help the business? What kind of software system?

- Yes, I think a system could help our business, perhaps a website would be beneficial to scale our company and reach out to more potential customers.

14. Are there any concerns in using a software system?

- My main concerns are the maintenance cost and the lack of an IT team to maintain and operate the website.

15. Do you have any preferred design that should be included in the user interface?

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- I don't have a definite idea of the design, maybe you could recommend a specific design or style.

16. Would you like features in the web system such as a 3D model where customers can visualize your products?

- That would be interesting to see and could help the customers more in visualizing our designs. I would also like to see a dashboard or an overview of our sales in the past months or so.

17. What are the steps from inquiry → quotation → approval → installation?

- From inquiry, product presentation, getting details of the stone specifications and measurements, suggestion on the design of the countertops such as edging details, provision of splash boards, fascia boards, sideboards, sometimes offer freebies or discounts, approval, fabrication or cutting at our shop or in some cases fabrication and cutting will be done on the job site before installation.

18. What part of your process takes the most time?

- The fabrication

19. Can quotations be revised? How many times?

- Yes, as many times as needed or required

20. Do you track quotation status?

- Yes, we do follow ups

21. Can you provide datasets up to 3 years or more?

- Yes, we can provide datasets up to 3 years.

Appendix B - Documentation

Interview with Six Sigmaphil



Sample Projects of Six Sigmaphil





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Appendix C Sample Dataset and Quotation

SIX SIGMAPHIL Enterprises Corporation

Namayan, Mandaluyong City

Phone Nos. 09195859959 / 824-53459

December 16, 2025

PROJECT : Narra Residence
Indang, Cavite 4122

STONE: ZIRCON GOLD PREMIUM QUARTZ

Kitchen Countertop w/Splashboard up to hanging cabinet

Material Cost	6.85	sq.m	@	11,800.00	/sq.m		80,830.00
Fabrication Cost							
Cutting	17.90	l.m	@	140.00	/l.m	2,506.00	
Polished Edge	2.82	l.m	@	700.00	/l.m	1,974.00	
Miter Joint	3.47	l.m	@	900.00	/l.m	3,123.00	7,603.00
Installation Cost							
	4.23	sq.m	@	1,900.00	/sq.m	8,037.00	<u>8,037.00</u>
						Subtotal	96,470.00

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Bar Countertop w/waterfall siding

Material Cost	3.18	sq.m	@	11,800.00	/sq.m	37,524.00
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Fabrication Cost

Cutting	7.02	1.m	@	140.00	/1.m	982.80
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Miter Joint	4.34	1.m	@	900.00	/1.m	3,906.00
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Installation Cost

	1.45	sq.m	@	1,900.00	/sq.m	2,755.00
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Subtotal	50,056.60
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Total Amount	146,526.60
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Less 5% discount	<u>(7,326.33)</u>
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139,200.27

Delivery Cost	8,500.00
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Mobilization Cost	9,500.00
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Net Amount	157,200.27
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Less special discount	<u>(27,200.27)</u>
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TOTAL CONTRACT AMOUNT	Php	130,000.00
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VAT NOT INCLUDED

PAYMENT TERMS:

- 70% of Total Contract Amount as down payment
- 30% balance upon completion of work

Should there be any queries, please do not hesitate to call the undersigned.

By:

Conforme:

Roland B. Koh

General Manager

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Appendix D Letter to Conduct Research

April 15, 2026

Kyle Lopez
Bachelor of Science in Information Technology
Jose Rizal University
Mandaluyong City
dreizenkyle.lopez@my.jru.edu
09690631147

Roland Koh
General Manager
Six Sigmaphil Enterprises Corporation
Brgy Namayan, Neptune st. Mandaluyong City, Metro Manila
1551

Dear Roland,

Greetings!

I hope this letter finds you well and in good spirits. I'm writing on behalf of a group of Information Technology students at Jose Rizal University. Our group is made up dedicated students who are eager to participate in a meaningful research project that meets both our academic requirements, specifically Project 1 - Thesis 1, and the needs of your esteemed organization.

Our research group includes the following members:

Christian Enayon – Programmer - 09928356117 – christian.enayon@my.jru.edu
Dreizen Kyle Lopez – Project Manager - 09690631147 – dreizenkyle.lopez@my.jru.edu
Keann Geoff Aguinaldo – Network Designer-09175978048 – keanngeoff.aguinaldo@my.jru.edu
Prince Lloyd Sazon – Systems Analyst - 09917796606 – princelloyd.sazon@my.jru.edu

With the guidance and approval of our subject adviser, Ronel Jaron Bilog, and our technical adviser, Ronel Jaron Bilog, we are eager to begin a research project that will include system development tailored to your organization's specific requirements and challenges. Our goal is to contribute to the improvement of processes, efficiency, and overall effectiveness by implementing innovative technological solutions.

Our proposed research plan includes, but is not limited to, the following activities:

1. Data Gathering: We intend to collect relevant data and information about your organization's current systems, workflows, and areas for improvement.

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2. Surveys: We intend to conduct surveys among stakeholders in your organization to gather insights, feedback, and suggestions that will help shape our system development strategy.

3. Research Activities: We will conduct rigorous analysis, design, development, testing, and evaluation phases to ensure the viability and effectiveness of the proposed system solutions.

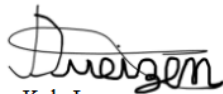
We are committed to maintaining the highest levels of professionalism, confidentiality, and integrity throughout the research process. Furthermore, we guarantee that our research activities will be carried out in strict accordance with ethical principles and all applicable organizational policies and regulations.

We would appreciate your cooperation and support in facilitating access to relevant resources, data, and personnel who can provide valuable insights and assistance during our research.

We believe that this collaborative effort will not only improve our learning experience, but will also provide tangible benefits to your organization. We are excited to contribute our skills, knowledge, and enthusiasm to this project and look forward to working closely with you.

Thank you for considering our proposal. If you have any questions or need more information, please do not hesitate to contact us at the following numbers or email addresses.

Yours sincerely,



Kyle Lopez
Project Manager for Primarchs

Noted by:



Ronel Jaron Bilog, DIT

Roland Koh

Appendices E Sample 3D Model from Meshy.AI

